

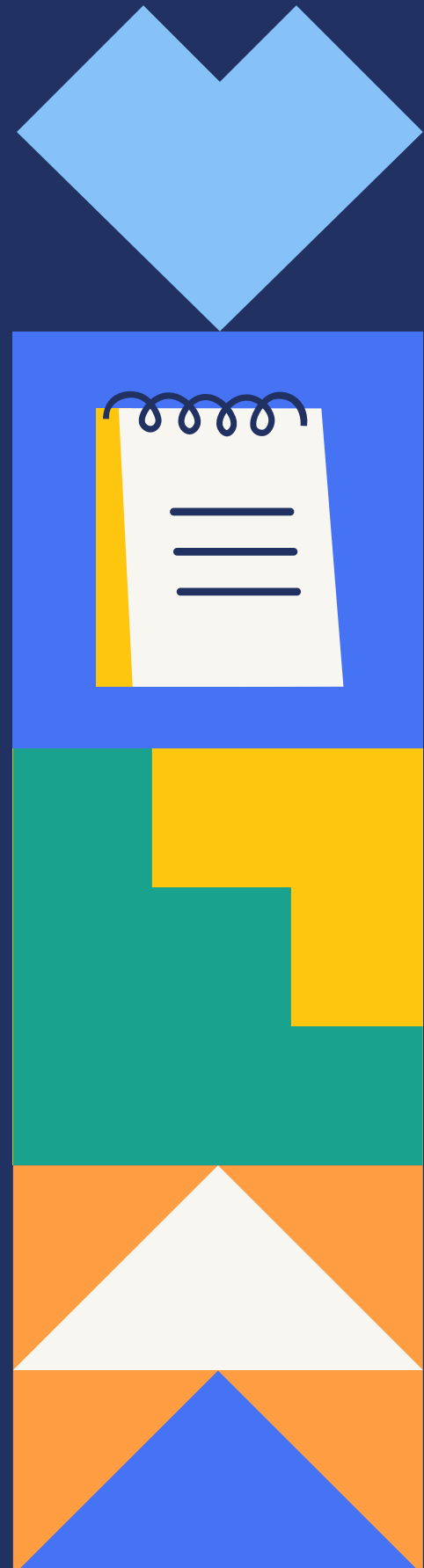
Do movies that
resonate with audiences
also earn stronger
critical reception?

Group: Root_R_1.2

IT3011 - Theory and Practices in Statistical Modelling



Group Details



Member Details

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Problem Statement

"Content that aligns with audience preferences receives better feedback."



- **Movie ratings are influenced by subjective factors (story, cast, individual taste) and critics do not always agree with fans.**
- **In this study, We Are investigating the alignment between audience and critic reception.**

Objective of Analysis

To statistically evaluate:

higher fan ratings are statistically associated with higher critic ratings

Using:

- Descriptive Analytics
- Inferential Analytics
- Predictive Analytics



Dataset Information

Dataset Used: Rotten Tomatoes Movies and Critic Reviews Dataset

Source: Kaggle

Type of Data : Secondary dataset

Records : 17713

Fields : 22

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
rotten_tomatoes_id	movie_title	movie_info_critics_content	genres	directors	authors	actors	original_release_date	streaming_release_date	runtime	production_company	tomato_meter	tomato_meter_critics	tomato_meter_audience	audience_score	audience_score_critics	audience_score_audience
m/0814255	Percy Jackson & the Lightning Thief	PG	Action & Adventure	Chris Columbus	Craig T. Nelson	Logan Lerman	2/12/2010	11/25/2015	119	20th Century Fox	79	79	79	79	79	79
m/0878835	Please Give Kate (Cath) Nicole Ho R	PG	Comedy	Nicole Ho	Nicole Ho	Catherine Keener	4/30/2010	9/4/2012	90	Sony Pictures	87	87	87	87	87	87
m/10	10 A success	PG	Comedy	F Blake Edwards	Blake Edwards	Dudley Moore	10/5/1979	7/24/2014	122	Warner Bros	67	67	67	67	67	67
m/1000013-12_angry	12 Angry Men	NR	Classics	Sidney Lumet	Reginald F. Martin	Baillie	4/13/1957	1/13/2017	95	Criterion	100	100	100	100	100	100
m/1000079-20000	20,000 Leagues Under the Sea	PG	Action & Adventure	Richard Fleischer	Earl Felton	James Mason	1/1/1954	6/10/2016	127	Disney	89	89	89	89	89	89
m/10000_bc	10,000 B.C. Mammoth	PG-13	Action & Adventure	Roland Emmerich	Harald Kloser	Steven Strait	3/7/2008	6/22/2013	109	Warner Bros	8	8	8	8	8	8
m/1000121-39_step	The 39 Steps	NR	Action & Adventure	Alfred Hitchcock	Alma Reville	Robert Donat	8/1/1935	1/12/2017	80	Gaumont	96	96	96	96	96	96
m/1000123-310_to	3:10 to Yuma	NR	Classics	Delmer Daves	Halsted Wright	Glenn Ford	8/7/1957	4/16/2012	92	Columbia	96	96	96	96	96	96
m/10002008-charly	Charly	PG	Comedy	Adam Rifkin	Thomas Jack	Wendy Hoffman	8/9/2002	5/22/2017	103	Excel Entertainment	20	20	20	20	20	20
m/10002004-abraham	Abraham Lincoln	NR	Classics	David W. Griffith	Gerrit J. Li	Walter Huston	11/8/1930	12/3/2013	97	United Artists	82	82	82	82	82	82
m/10002114-dark_w	Dark Water	PG-13	Art House	Hideo Nakata	Hideo Nakata	Hitomi Ku	1/19/2002	3/23/2017	100	Toho	60	60	60	60	60	60
m/1000224-accuse	The Accused	R	Drama	Jonathan Demme	Tom Toporoff	Jodie Foster	#####	10/4/2016	110	Paramount	91	91	91	91	91	91
m/10002516-lost_ci	The Lost City	R	Drama	Andy Garcia	G. Cabrera	Andy Garcia	9/3/2005	3/23/2017	143	Magnolia	25	25	25	25	25	25
m/10002519-breaki	The Break	NR	Drama	Michael C. Rinaldi	Michael C. Rinaldi	John Garfield	10/6/1950	8/25/2016	97	Warner Bros	100	100	100	100	100	100
m/1000253-adams	Adam's Rib	NR	Classics	George Cukor	Garson Kanin	Spencer Tracy	#####	5/1/2008	101	MGM Home	96	96	96	96	96	96
m/10002635-bridge	The Bridge	PG	Art House	Mary McGarry	Mary McGarry	Gabriel Byrne	6/10/2005	7/24/2014	124	Fine Line	4	4	4	4	4	4
m/10002673-growl	The Growl	PG	Drama	M. Joseph Lo	Robert Thibault	Van Heflin	12/1/1951	12/5/2016	92	VCI Entertainment	100	100	100	100	100	100
m/10003276-crimin	Criminal	PG	Drama	M. Gregory	J. Gregory	John C. Reilly	9/24/2004	4/27/2016	87	Warner Bros	69	69	69	69	69	69
m/1000334-adventu	The Advent	PG	Action & Adventure	Will Vinton	Susan Shuler	James Whit	3/1/1985	2/4/2017	86	Eureka	80	80	80	80	80	80
m/10003437-deep_1	Deep Blue	PG	Documentary	Andy Byatt	Alastair Fyfe	Pierce Brosnan	6/17/2005	12/7/2016	90	Miramax	67	67	67	67	67	67
m/1000355-adventu	The Advent	PG	Action & Adventure	Michael C. Norman	R. Errol Flynn	5/14/1938	1/23/2014	102	Warner Bros	100	100	100	100	100	100	100
m/10003876-man_h	Man Hunt	NR	Classics	Fritz Lang	Dudley Ni	Walter Pidgeon	1/1/1941	3/1/2014	92	Twentieth Century Fox	75	75	75	75	75	75
m/10003925-dead_x	Dead End	NR	Comedy	Jean-Baptiste	Jean-Baptiste	Ray Wise	9/26/2003	12/11/2018	83	Lions Gate	75	75	75	75	75	75
m/10004050-bandw	Bandwagon	NR	Comedy	John Schu	John Schu	Matthew Perry	9/12/1997	10/3/2016	103	Avalanche	80	80	80	80	80	80
m/10004069-whore	Whore	NR	Art House	Marisa Li	Adela Iba	Daryl Han	4/16/2004	2/4/2014	77	Dolores	0	0	0	0	0	0
m/10004209-tristan	Tristan & Isolde	PG-13	Drama	Kevin Reynolds	Dean Cain	James Franco	1/13/2006	11/25/2015	116	20th Century Fox	31	31	31	31	31	31
m/10004226-wild_s	Wild Side	NR	Art House	Sébastien	Sébastien	Stéphane	6/10/2005	10/25/2005	94	Wellspring	63	63	63	63	63	63
m/10004232-100042	A Tale of Two Cities	PG	Comedy	Vincent S.	Vincent S.	Vincent P.	11/7/2003	1/16/2018	82	Rotten	38	38	38	38	38	38
m/10004288-runni	Running	R	Action & Adventure	Wayne K.	Wayne K.	Paul Walker	2/24/2006	10/1/2008	121	New Line	41	41	41	41	41	41
m/10004352-going	Going the Distance	R	Comedy	Mark Griffin	Eric Good	Christopher Ja	8/20/2004	3/10/2017	93	Saville	0	0	0	0	0	0
m/10004354-ultravi	Ultraviolet	PG-13	Action & Adventure	Kurt Wimmer	Kurt Wimmer	Milla Jovovich	3/3/2006	4/16/2012	88	Sony Pictures	8	8	8	8	8	8
m/10004635-home	Home of the Brave	NR	Action & Adventure	Paola di F.	Paola di F.	Mary McCormack	5/11/2004	8/1/2016	75	Emerging	93	93	93	93	93	93



Data Preprocessing

Performed:

- Removed duplicates
- Removed missing values
- Validated rating ranges (0–100)
- Removed outliers
- Added derived fields

Final Sample: 15,165 movies retained after cleaning (14.4% loss)

Final cleaned dataset prepared for analysis.



Descriptive Analysis Methods

Used

- Summary statistics
- Histograms
- Boxplots
- Trend charts
- Correlation heatmap

To understand rating distributions and audience behavior.



Summary Statistics

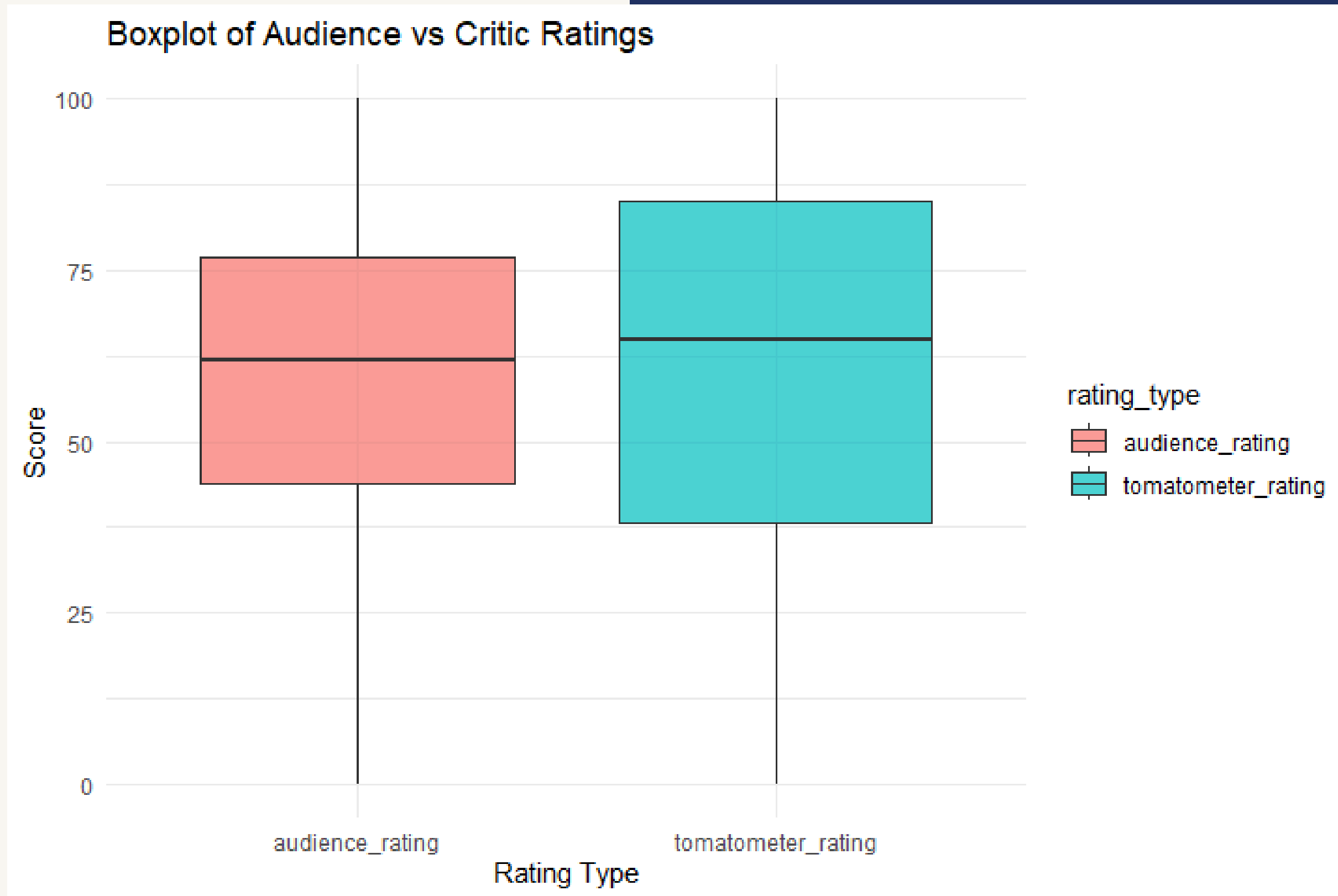
```
Summary of Tomatometer Rating:
> print(summary(movies_clean$tomatometer_rating))
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00   38.00   65.00   60.03   85.00  100.00
> cat("Standard Deviation:", round(sd(movies_clean$tomatometer_rating, na.rm = TRUE), 3), "\n")
Standard Deviation: 28.458
>
> cat("\nSummary of Audience Rating:\n")

Summary of Audience Rating:
> print(summary(movies_clean$audience_rating))
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00   44.00   62.00   60.28   77.00  100.00
> cat("Standard Deviation:", round(sd(movies_clean$audience_rating, na.rm = TRUE), 3), "\n")
Standard Deviation: 20.394
> |
```

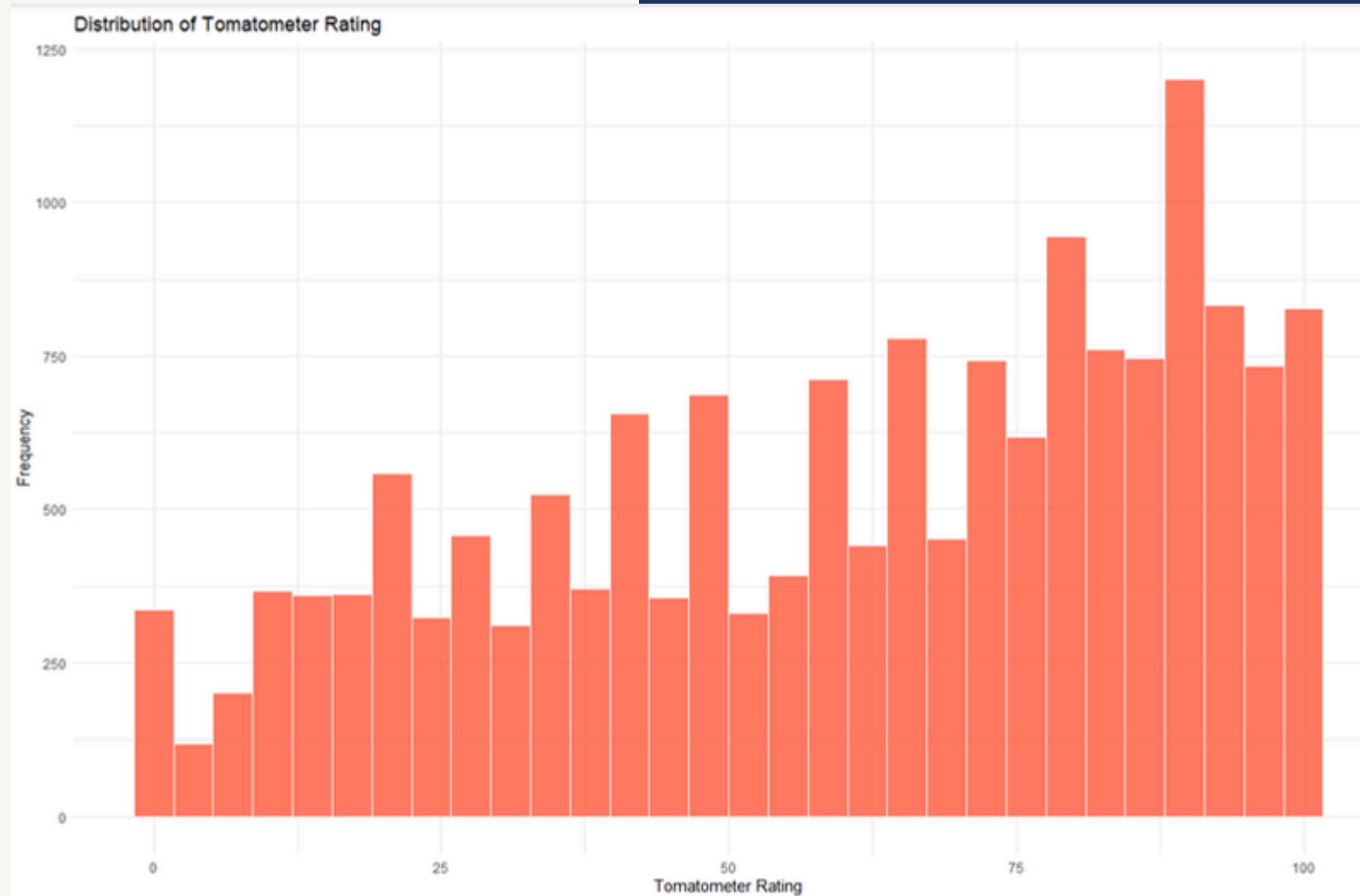
Insights :

- Critics are more polarizing than audiences
- Both groups average the same

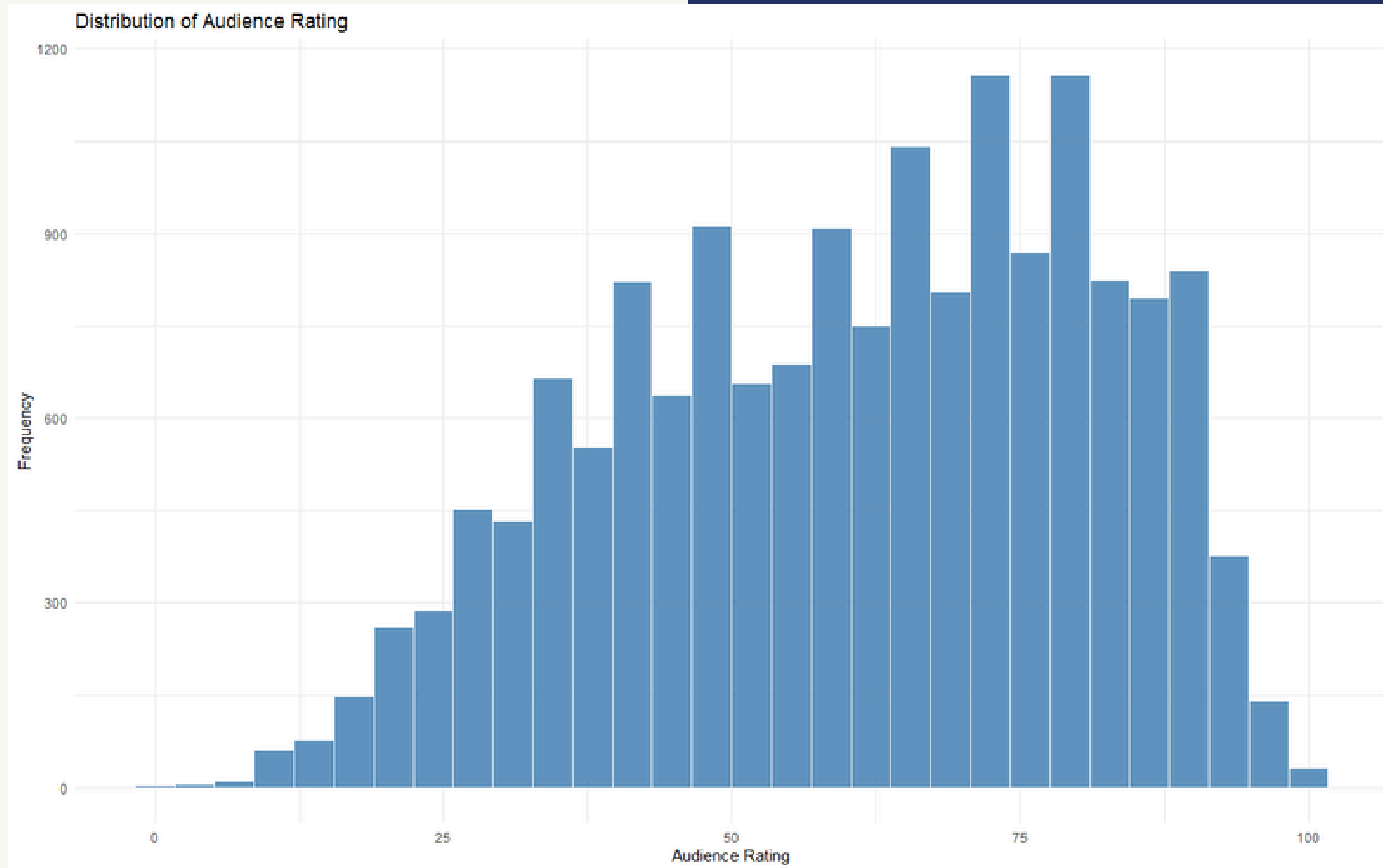
boxplot of audience and tomatometer rating



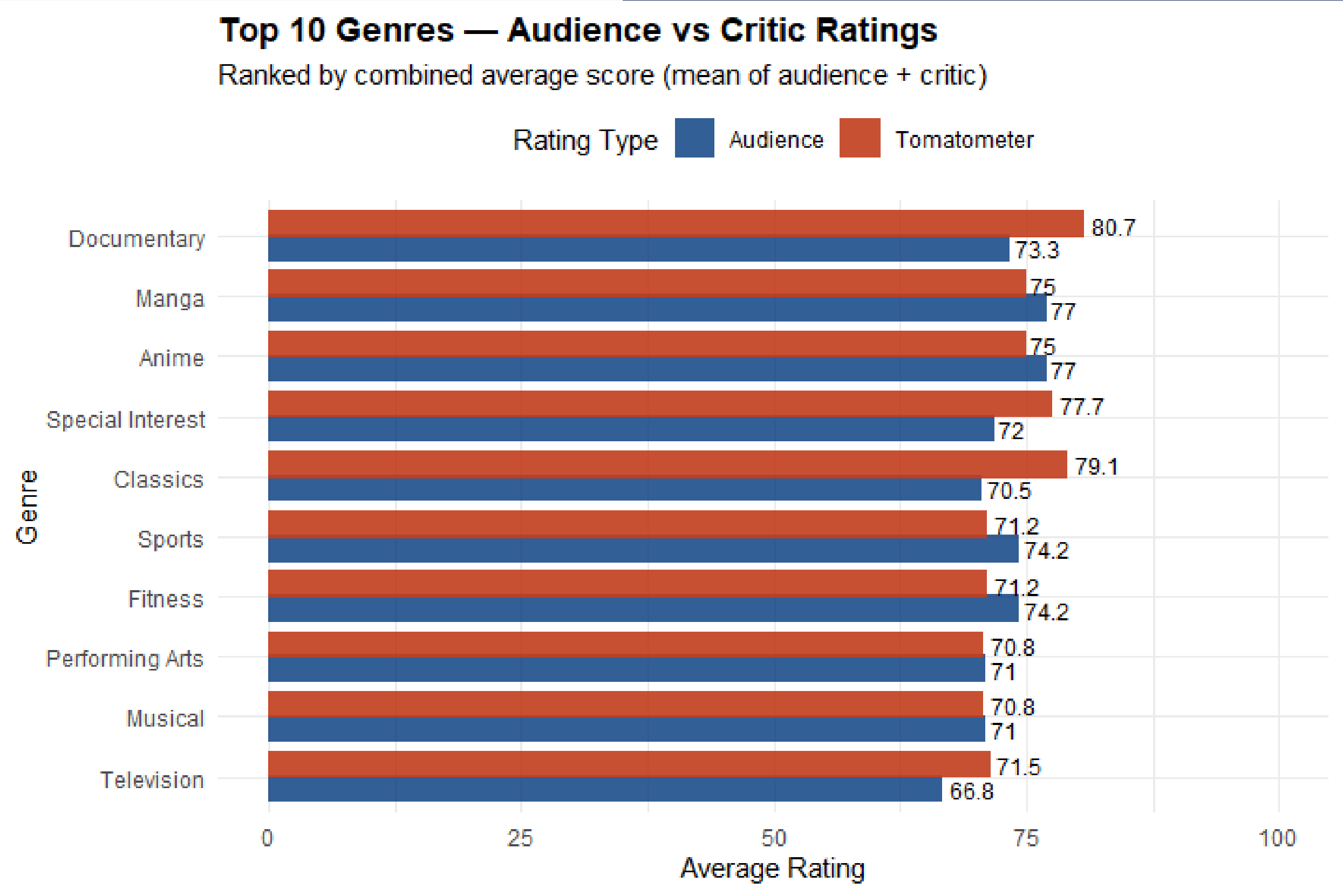
Distribution of Tomatometer Rating



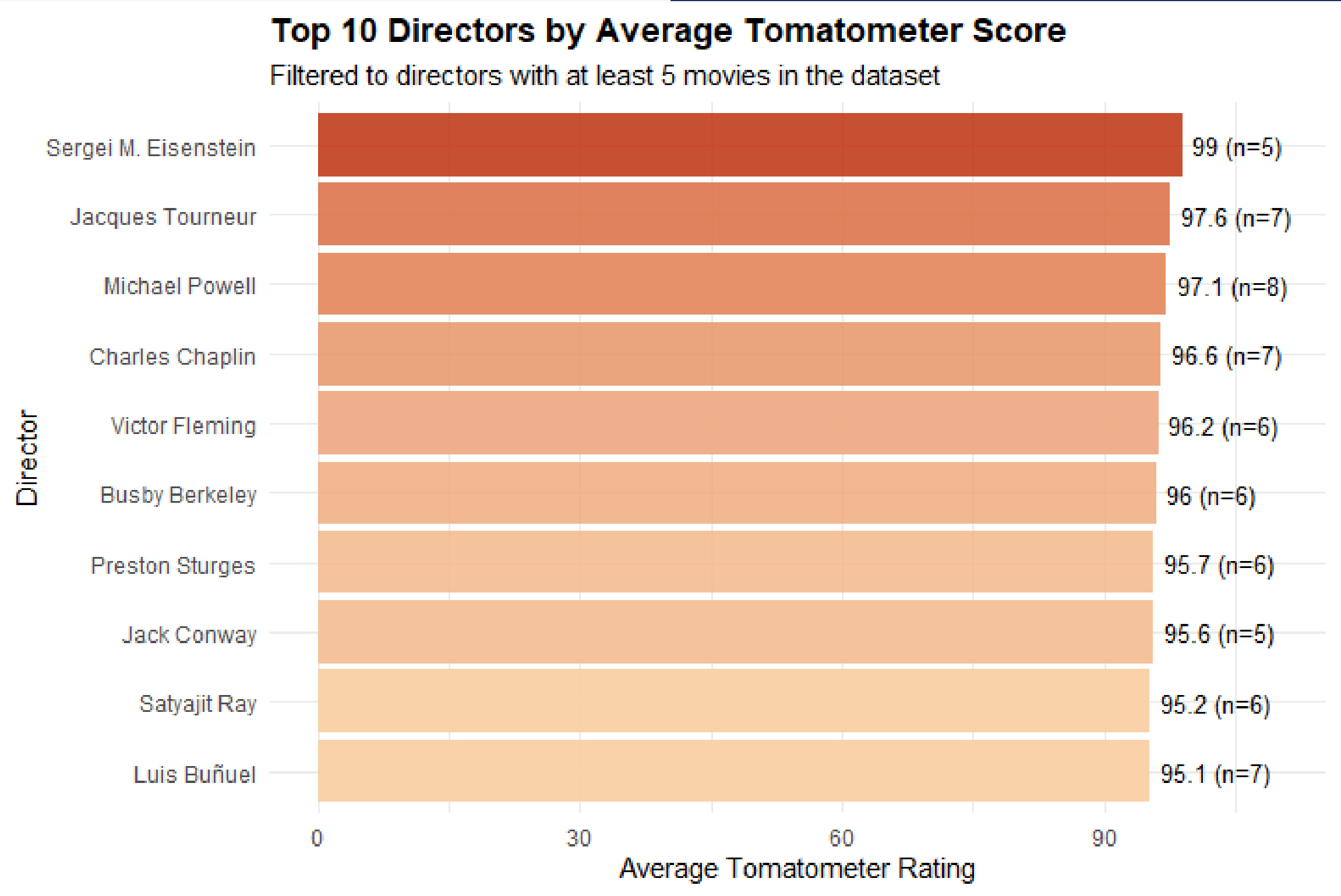
Distribution of Audience Rating



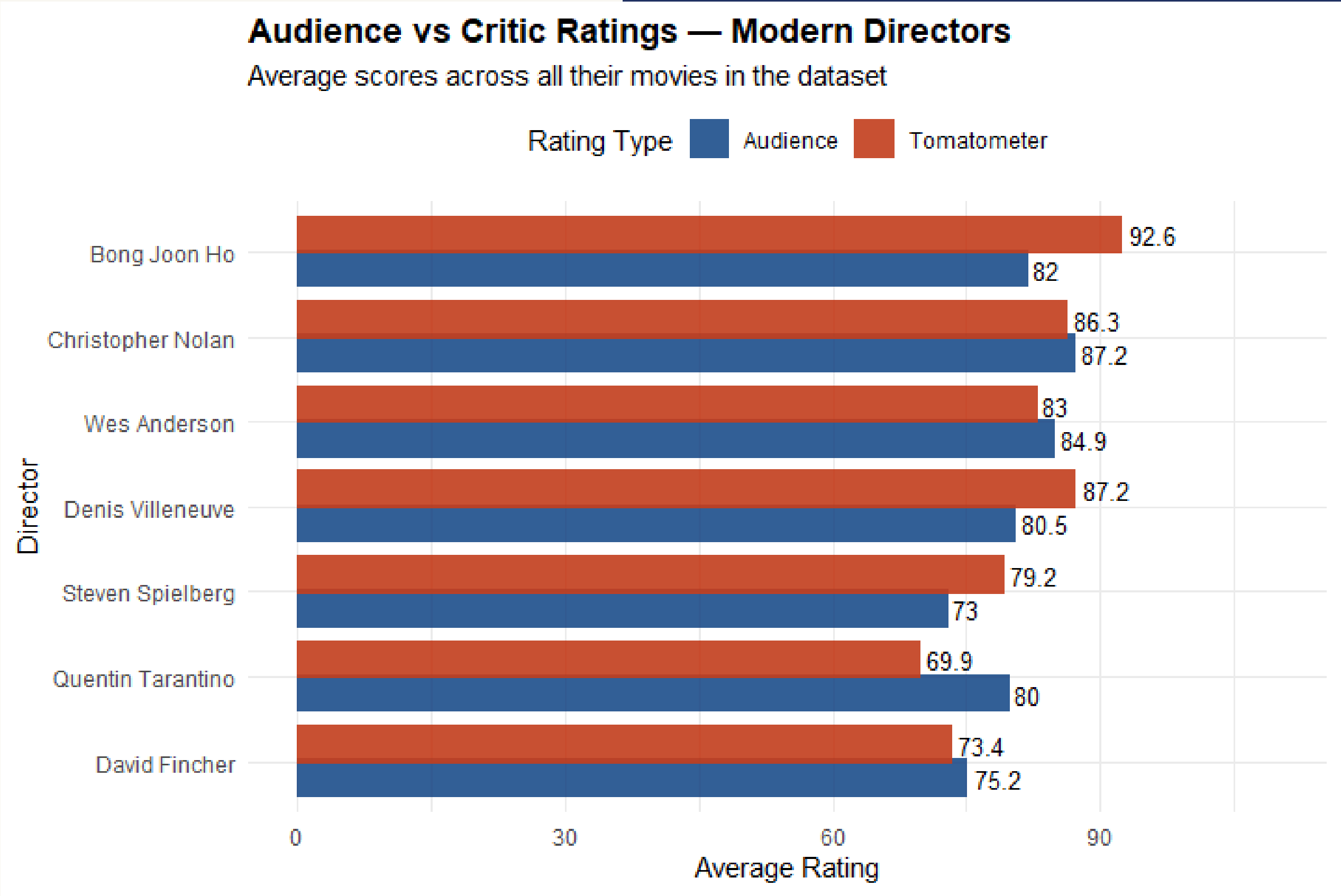
Audience vs Critics Ratings



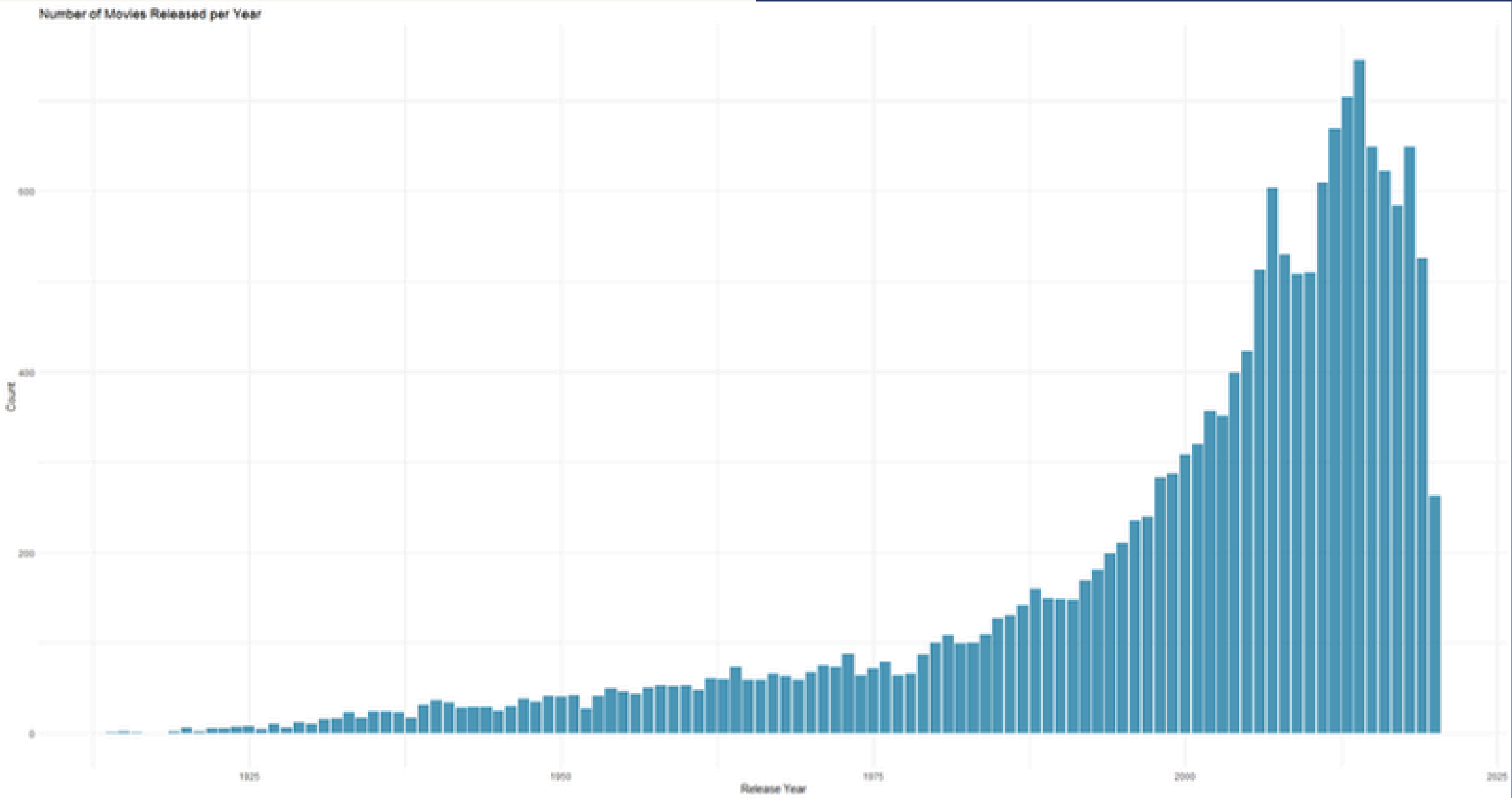
Top 10 Directors by Average Tomatometer Score



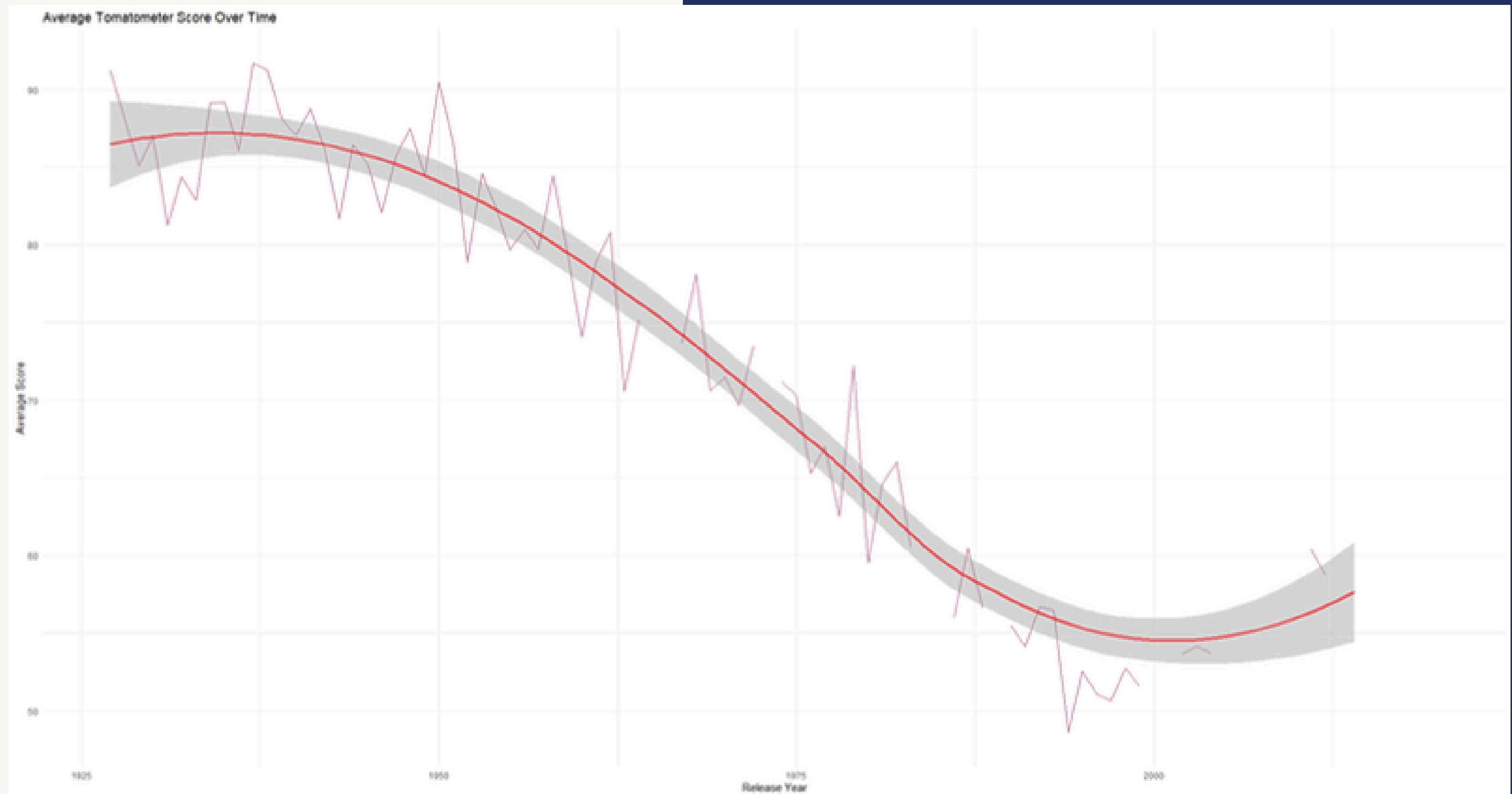
Modern Directors — Average Audience vs Critic Ratings



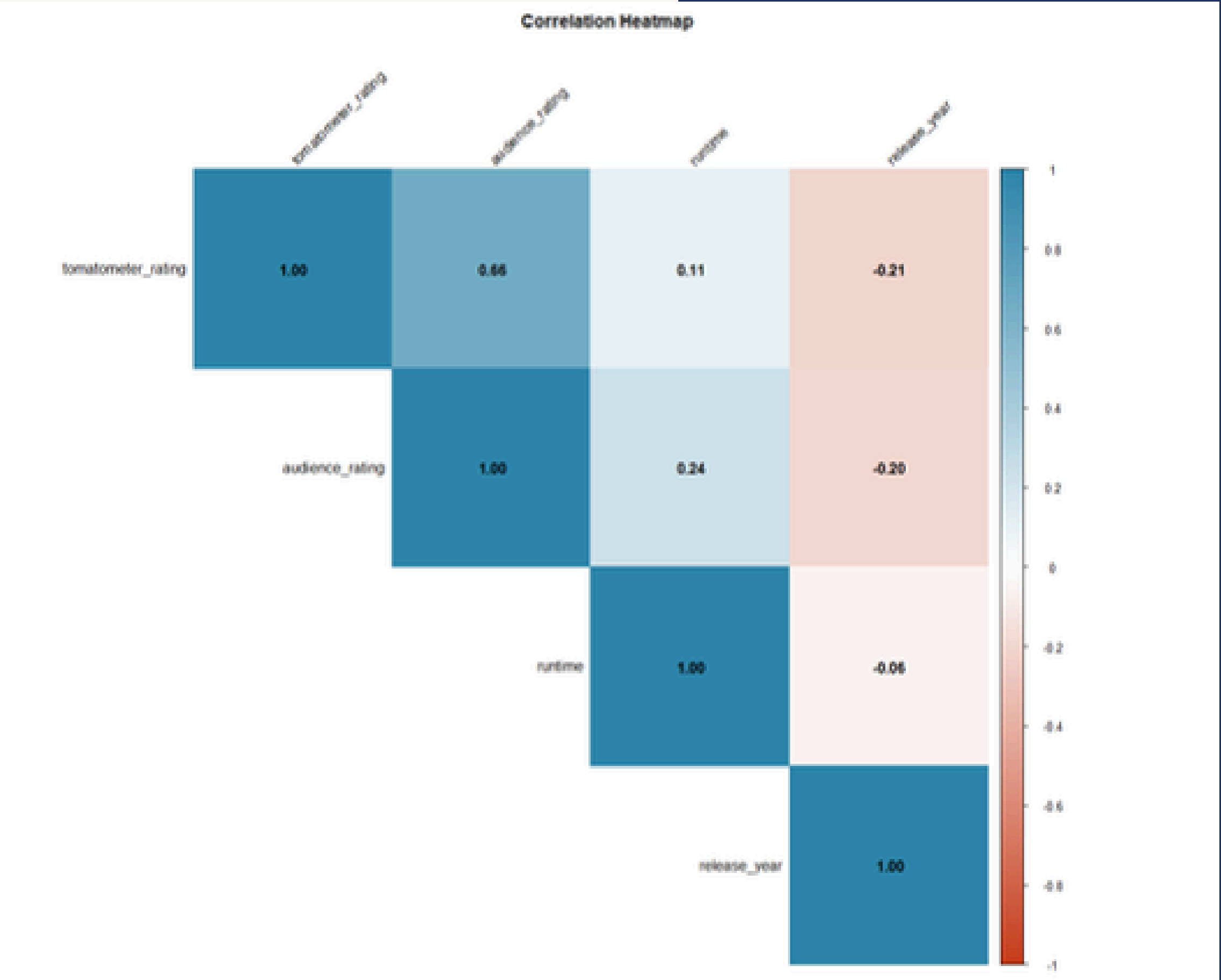
Movies released per year



Average Tomatometer over time



Correlation heatmap of numeric features



Descriptive Results

Key Findings

- ✓ Audience ratings are generally higher than critic ratings
- ✓ Certain genres attract stronger audience support
- ✓ Ratings vary across content categories
- ✓ Positive relationship visible between audience and critic ratings



Inferential Analysis

Objective

To test whether audience preferences significantly influence feedback scores.

Hypothesis

H_0 : No relationship exists between audience rating and critic rating

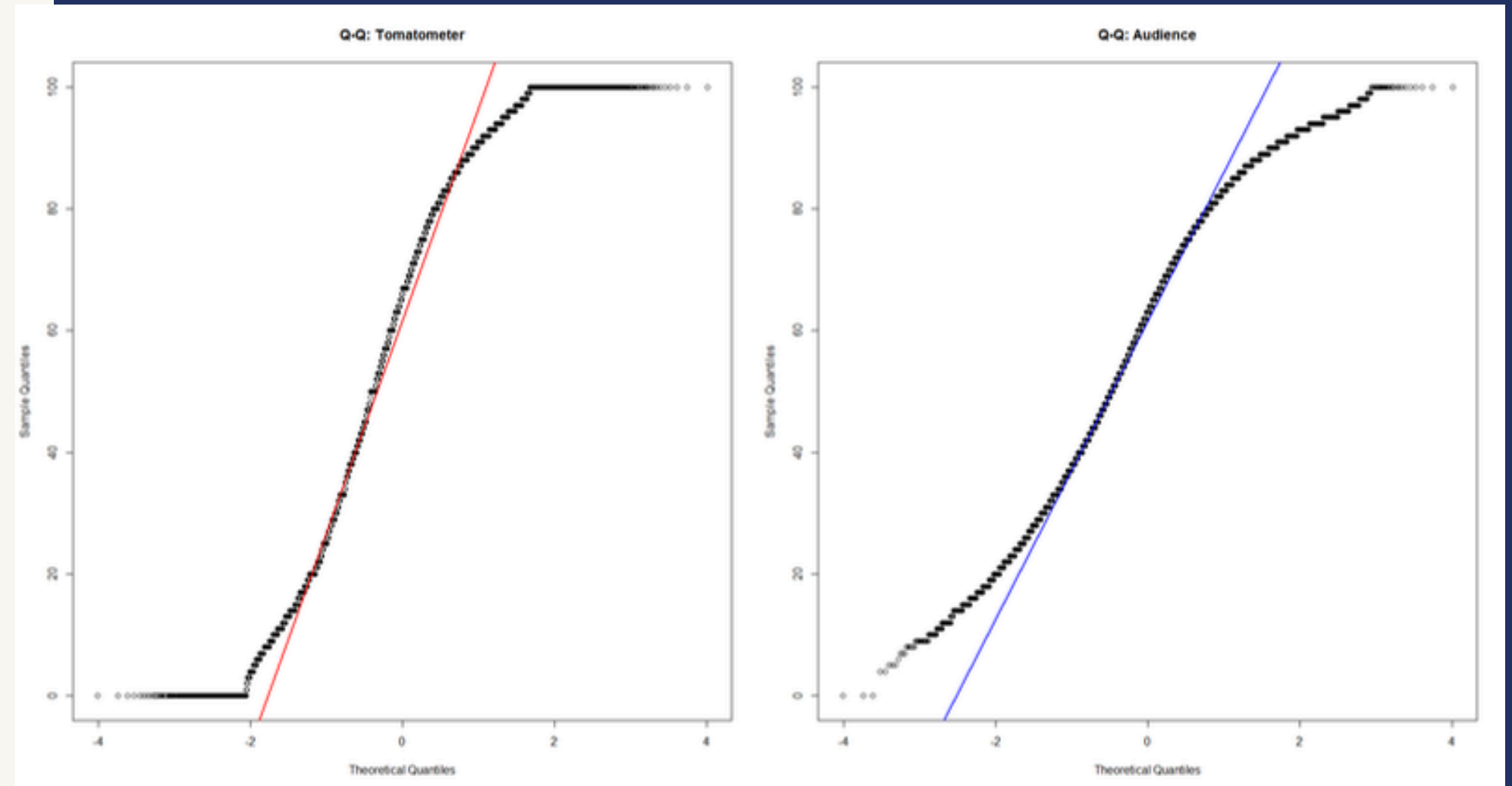
H_1 : Significant positive relationship between audience rating and critic rating



Normality Assessment

```
> safe_shapiro(movies_clean$tomatometer_rating, "tomatometer_rating")
tomatometer_rating      W = 0.9392, p = 3.353e-41
> safe_shapiro(movies_clean$audience_rating, "audience_rating")
audience_rating        W = 0.9720, p = 3.45e-30
>
```

- Shapiro-Wilk Test Normality assumption is rejected
- Indicates data is not normally distributed
- Q-Q plots show clear deviation from straight line
- Indicates data is not normally distributed



Correlation Test

Spearman correlation

Spearman's ρ

0.666

$p < 2.2e-16$, Strong positive monotonic relationship



- Data is not normally distributed
- Used Spearman correlation (non-parametric)
- Moderate to strong positive relationship

spearman's rank correlation rho

```
data: movies_clean$audience_rating and movies_clean$tomatometer_rating
s = 2.4262e+11, p-value < 2.2e-16
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.6677987
```

Inferential Results

Correlation Output

- Positive correlation coefficient
- $p\text{-value} < 0.05$

Decision

Reject H_0

Interpretation

Movies receiving stronger audience approval tend to receive better critic feedback.



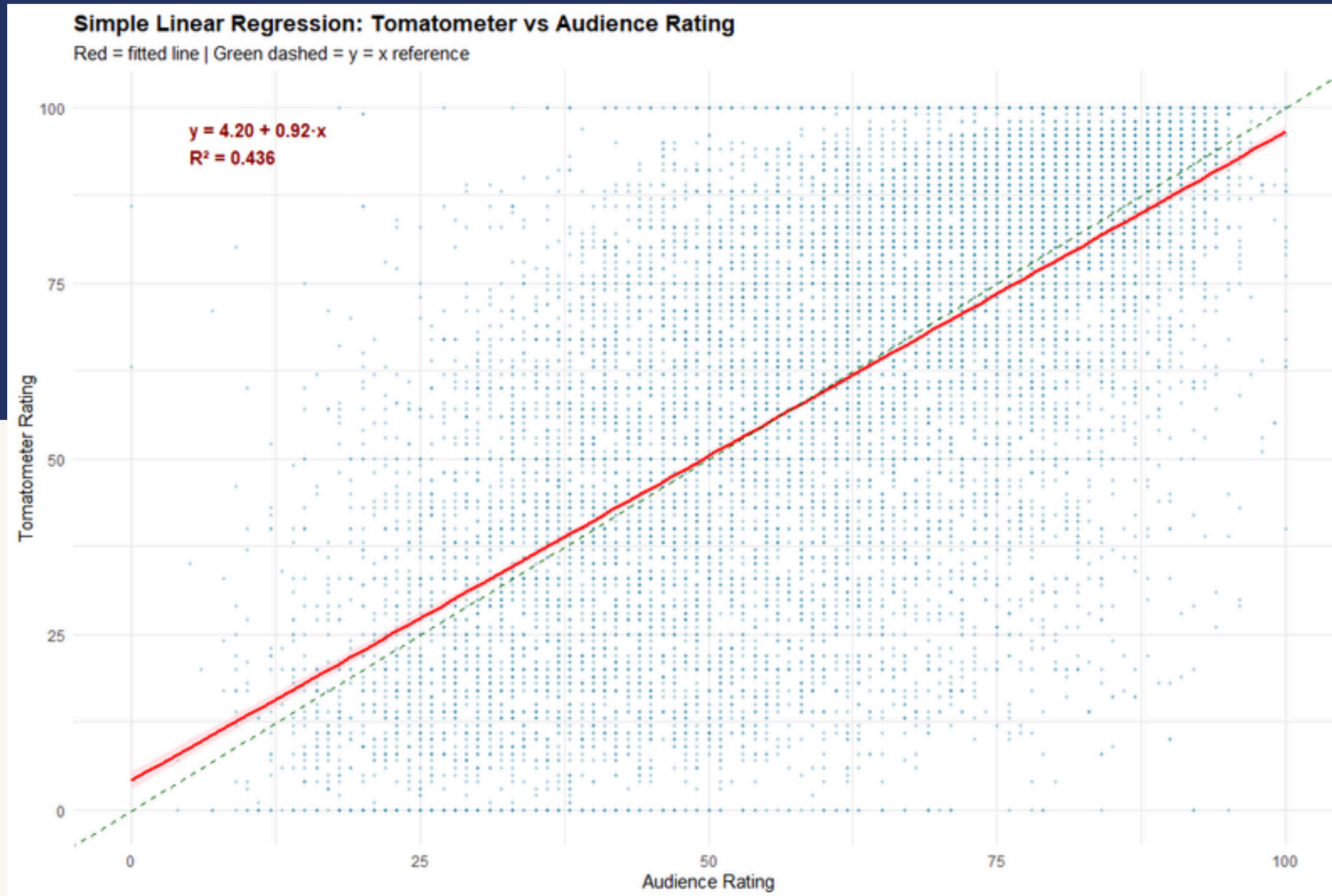
Predictive Analytics

Models Developed

- **Simple Linear Regression**
- **Multiple Linear Regression**
- **Random Forest**
- **Gradient Boosting**
- **XGBoost**



Simple Linear Regression



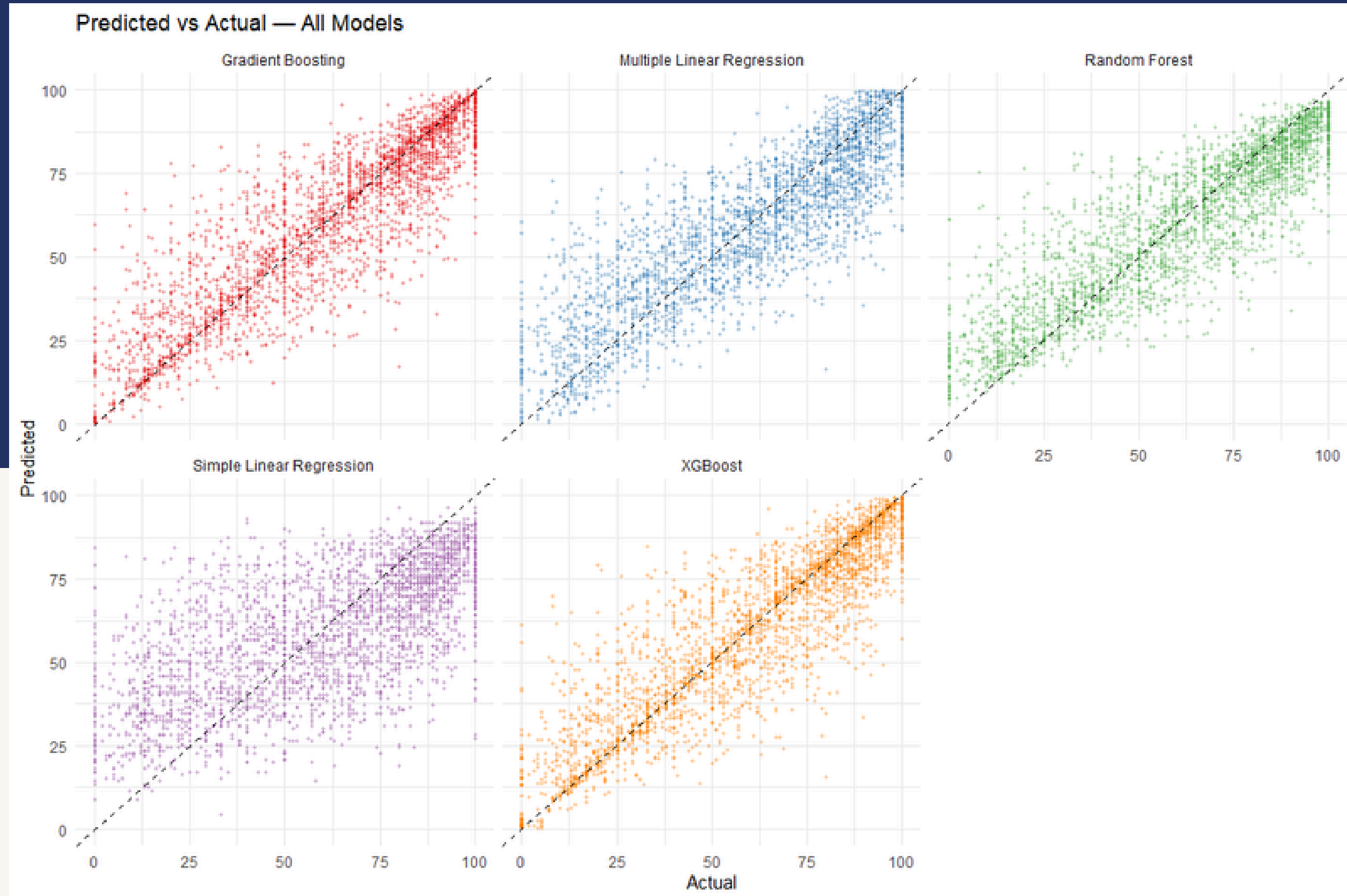
Regression equation:

$$\text{tomatometer} = 4.203 + 0.923 \times \text{audience_rating}$$

R-squared :

0.4359 (43.6% variance explained)

All Models



Model Performance

Model	RMSE	MAE	R ²
Linear Regression	21.384	17.096	0.432
Multiple LR	14.047	10.669	0.755
Random Forest	13.541	10.024	0.778
Gradient Boosting	12.917	8.936	0.792
XGBoost	12.834	8.748	0.795

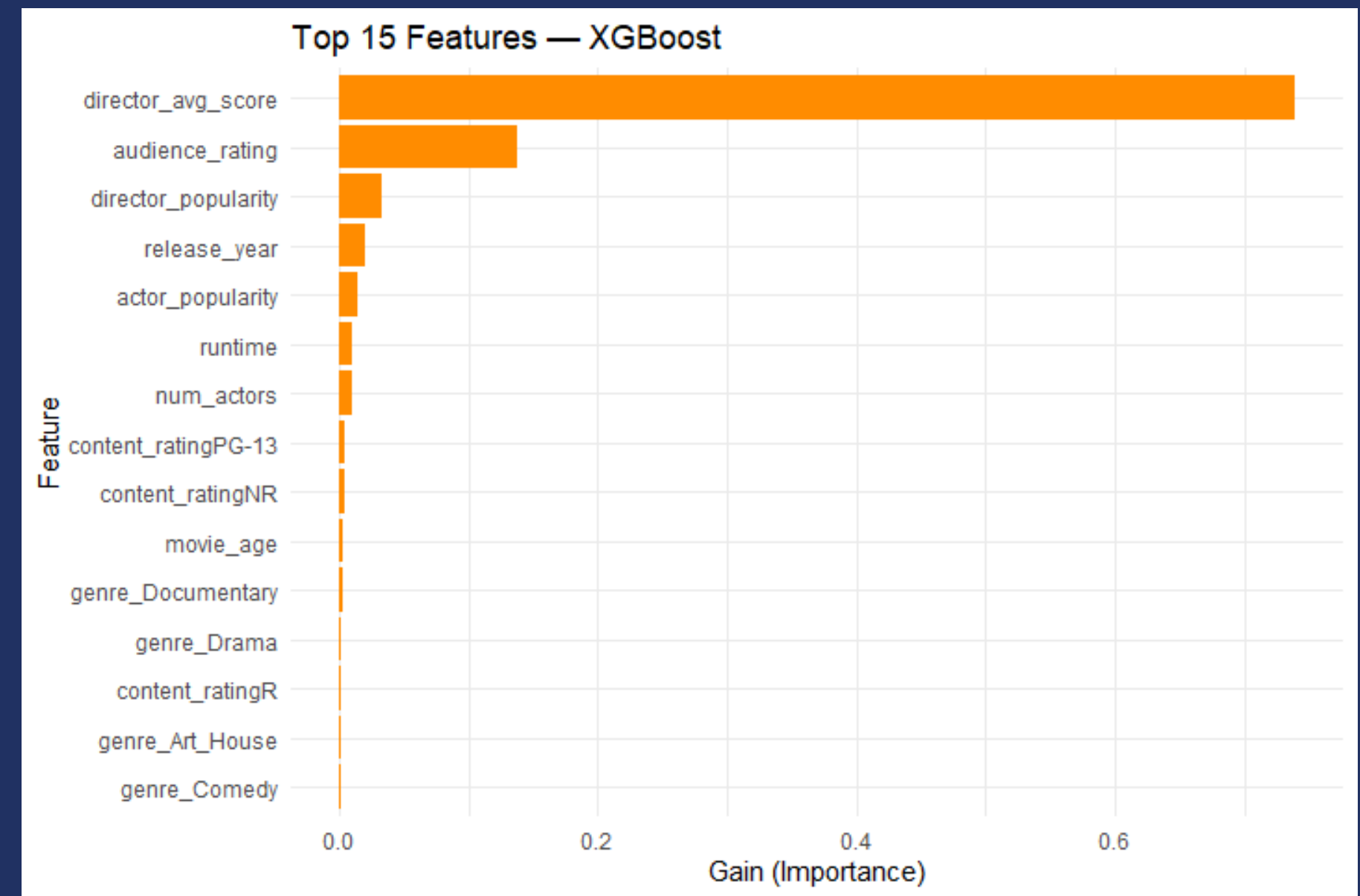


Best Model

Best Performing Model: XGBoost

Why?

- Lowest prediction error
- Highest explanatory power
- Best test accuracy



Final Interpretation

Audience preference is a meaningful predictor of movie feedback.

However, feedback is also influenced by:

- Genre quality
- Storyline
- Runtime
- Direction & acting



Conclusion

The statistical evidence supports the statement:

Content that aligns with audience preferences receives better feedback.



THANK YOU

Canva